Year 7
Mathematics

Unit 3: Equations, Functions and Formulae

3.4 Writing Formulae

Writing Formulae

STARTER · 5 MIN

Date: _____ **Title: Writing Formulae**

LEARNING OBJECTIVES

- To be able to write a formula from a description

STARTER *Answer in your book. No calculators.*

1. Write 5 more than x
2. Write 3 times n
3. $C = 4n$. Find C when $n = 6$
4. Write t divided by 2

Starter — Answers

STARTER

Tick or correct in green pen

1. $x + 5$

2. $3n$

3. $C = 24$

4. $\frac{t}{2}$

Success Criteria

SUCCESS CRITERIA

By the end of this lesson:

BRONZE	I can write a one-step formula from a description	☐
SILVER	I can write a formula with a fixed charge and a rate	☐
GOLD	I can write and use formulae to solve problems	☐

Tick each box in the plenary when you can do it on your own.

Introduction — Writing Formulae

INTRODUCTION · 3 MIN

A formula starts with the letter for the answer, then =, then an expression.

Choose letters

Say what each letter stands for.

C = cost, n = weeks

Rate

Amount each time × how many.

\$12 a week: $12n$

Fixed charge

Add it once.

\$5 plus \$3 a km: $5 + 3k$

Formula

Subject = expression.

$C = 12n$

Examples 1 and 2 — One-Step Formulae

EXAMPLES · 5 MIN

BRONZE

Watch and copy into your book

Hint Formula: C = ... Rate × amount

Example 1

Storage costs $12 per week.

Write a formula for the cost C for w weeks.

Example 2

A bike costs £8 per hour.

Write a formula for the cost C for h hours.

Examples 1 and 2 — Corrections

CORRECTIONS

BRONZE

Check your copy and correct in green pen

Example 1

Cost each week × number of weeks

$C = 12w$

Tip: check: 2 weeks gives $24.

Example 2

Cost each hour × number of hours

$C = 8h$

Tip: a formula needs C = ...

Task 1

PRACTICE · 9 MIN

BRONZE *Write a formula*

Hint Formula: C = ... Rate × amount

1. Pens cost 30p each.
   Cost C of n pens

2. Tickets cost $9 each.
   Cost T of t tickets

3. 5 sweets in each bag.
   Sweets S in b bags

4. 60 minutes in an hour.
   Minutes M in h hours

5. Share p pizzas between 4.
   Each gets E pizzas

★ Ali is 3 years older than Sam.
Ali's age A, Sam's age s

Task 1 — Answers

CORRECTIONS

BRONZE

Tick or correct in green pen

1 $C = 30n$

2 $T = 9t$

3 $S = 5b$

4 $M = 60h$

5 $E = \frac{p}{4}$

★ $A = s + 3$

Examples 3 and 4 — Fixed Charge plus a Rate

EXAMPLES · 5 MIN

SILVER

Watch and copy into your book

Hint Fixed charge + rate × amount Multiply before adding

Example 3

A phone plan costs $12.50 plus $4 per GB of data.

Write a formula for the cost C for n GB.

Example 4

Cooking a chicken takes 45 min per kg plus 20 min.

Write a formula for the time T for k kg.

Examples 3 and 4 — Corrections

CORRECTIONS

SILVER

Check your copy and correct in green pen

Example 3

Fixed part + rate × amount

$$C = 12.50 + 4n$$

Tip: the $12.50 is paid once.

Example 4

Rate × amount + fixed part

$$T = 45k + 20$$

Tip: check: 2 kg gives 110 minutes.

Task 2

PRACTICE · 9 MIN

SILVER *Write a formula, then use it*

Hint Fixed charge + rate × amount Multiply before adding

1. Taxi: \$3 plus \$2 per km. Cost C for k km

2. Gym: \$20 + \$15 a month. Cost C for m months

3. Use your taxi formula for 8 km.

4. Use your gym formula for 6 months.

5. Hire: \$50 + \$25 per day. Cost H for d days

★ Use your hire formula for 3 days.

Task 2 — Answers

CORRECTIONS

SILVER

Tick or correct in green pen

1. $C = 3 + 2k$

2. $C = 20 + 15m$

3. $3 + 2 \times 8 = \$19$

4. $20 + 15 \times 6 = \$110$

5. $H = 50 + 25d$

★ $50 + 25 \times 3 = \$125$

Examples 5 and 6 — Formulae in Problems

EXAMPLES · 5 MIN

GOLD

Exam-style: show all your working

Hint Work backwards: inverses Mean = total ÷ how many

Example 5

(2 marks)

The mean M of three numbers a, b and c.

Write a formula for M.

Example 6

(3 marks)

A taxi costs \$4 plus \$3 per km.

(a) Write a formula.

(b) A trip costs \$31. How far?

Examples 5 and 6 — Corrections

CORRECTIONS

GOLD

Check your copy and correct in green pen

Example 5

Add them, then divide by how many

$$M = \frac{a + b + c}{3}$$

Tip: the fraction line works as a bracket.

Example 6

Write the formula, then work backwards

(a) $C = 4 + 3k$

(b) $31 - 4 = 27$

$27 \div 3$ = 9 km

Tip: undo + 4, then undo × 3.

Task 3

PRACTICE · 9 MIN

GOLD

Exam-style: write and use formulae

Hint Work backwards: inverses
Mean = total ÷ how many

1. Mean M of x and y

2. Perimeter P of a square of side s

3. Taxi: \$4 + \$3 per km.
Cost for 12 km?

4. Taxi: \$4 + \$3 per km.
A trip costs \$40. How far?

5. Hall: \$100 + \$8 per guest.
Formula for cost C, g guests

★ Hall: the cost is \$340.
How many guests?

Task 3 — Answers

CORRECTIONS

GOLD

Tick or correct in green pen

1 $M = \frac{x + y}{2}$

2 $P = 4s$

3 $4 + 3 \times 12 = \$40$

4 $(40 - 4) \div 3 = 12$ km

5 $C = 100 + 8g$

★ $(340 - 100) \div 8 = 30$

Plenary — Exit Ticket

PLENARY · 5 MIN

Answer in your book. Try all three.

BRONZE Tickets cost $7 each. Formula for the cost C of n tickets?

SILVER Taxi: $5 plus $2 per km. Formula for the cost C of k km?

GOLD Use your taxi formula to find the cost of 10 km.

Reflect: Look back at the learning objectives. Which ones can you do now?

Plenary — Answers

PLENARY

Tick or correct in green pen

BRONZE $C = 7n$

SILVER $C = 5 + 2k$

GOLD $C = 5 + 2 \times 10$

$= \$25$

Exam Practice — Writing Formulae

EXAM PRACTICE · 10 MIN

BRONZE

IGCSE exam-style: show all your working

Hint Formula: C = ... Rate × amount

Question 1

(1 mark)

Apples cost 25 cents each.

Write a formula for the cost, C cents, of n apples.

.....

Question 2

(2 marks)

There are 7 days in a week.

(a) Formula for D days in w weeks (1)

.....

(b) Use it to find D when $w = 9$. (1)

.....

Exam Practice — Fixed Charge and Rate

EXAM PRACTICE · 10 MIN

SILVER

IGCSE exam-style: show all your working

Hint
Fixed charge + rate × amount
Multiply before adding

Question 3

(2 marks)

A taxi charges $4 plus $3 for each kilometre.

Write a formula for the cost, C, of a journey of k kilometres.

.....

Question 4

(2 marks)

A plumber charges $30 plus $20 per hour.

Work out the cost of a 4-hour job.

.....

Exam Practice — Using Formulae

EXAM PRACTICE · 10 MIN

GOLD

IGCSE exam-style: show all your working

Hint Work backwards: inverses
Mean = total ÷ how many

Question 5

(3 marks)

A taxi charges \$4 plus \$3 for each kilometre.

(a) Formula for the cost C of k km (1)

.....

(b) A trip costs \$34. How far? (2)

..... km

Question 6

(2 marks)

Write a formula for the mean, M, of four numbers a, b, c and d.

.....

Exam Practice — Mark Scheme

CORRECTIONS

BRONZE

Mark your work: M = method mark, A = accuracy mark, B = independent mark

Question 1

$C = 25n$ **B1**

Tip: start with C =.

Question 2

(a) $D = 7w$ **B1**

(b) $D = 7 \times 9 = 63$ **B1**

Tip: check with 1 week = 7 days.

Exam Practice — Mark Scheme

CORRECTIONS

SILVER

Mark your work: M = method mark, A = accuracy mark, B = independent mark

Question 3

3k for the distance — **M1**

$C = 4 + 3k$ — **A1**

Tip: the $4 is paid once.

Question 4

$30 + 20 \times 4$ $= 30 + 80$ — **M1**

$= \$110$ — **A1**

Tip: multiply first.

Exam Practice — Mark Scheme

CORRECTIONS

GOLD

Mark your work: M = method mark, A = accuracy mark, B = independent mark

Question 5

(a) $C = 4 + 3k$ — B1

(b) $34 - 4 = 30$ — M1

$30 \div 3 = 10$ km — A1

Tip: work backwards.

Question 6

Add the four, divide by 4 — M1

$M = \dfrac{a + b + c + d}{4}$ — A1

Tip: divide the whole sum.